طرق التكامل **Method of Integration**

(I) Integration by substitution:

Consider the following integral

$$\int f(x)\,dx \longrightarrow (1)$$

Somtimes, we suggests a substitution for variable of integration of the form

$$y = g(x).$$

where $g(x)$ can be integrated.

Note: In general. We substitute the part of the integrand that is causing the difficulty.

Ex. Evaluate $\int \frac{\sqrt{x+2}}{x+6}\,dx$

Sol:

let $x+2 = y^2 \longrightarrow dx = 2y\,dy$

$$\therefore \int \frac{\sqrt{x+2}}{x+6}\,dx = \int \frac{y}{y^2+4} \cdot 2y\,dy = 2\int \frac{y^2+4-4}{y^2+4}\,dy$$

$$= 2\int \frac{(y^2+4-4)}{y^2+4}\,dy = \int 2\,dy - 8\int \frac{1}{y^2+4}\,dy$$

$$= 2y - 4\tan^{-1}\frac{y}{2} + c$$

$$= 2\sqrt{x+2} - 4\tan^{-1}\frac{\sqrt{x+2}}{2} + c \quad \#$$

Problems

* Evalute the following integrals.

(1) $\int \frac{x^2\,dx}{\sqrt{x+5}}$ (Hint $x+5 = y^2$)

(2) $\int \frac{dx}{1-\sqrt[3]{x+1}}$ (let $x+1 = t^3$)

(3) $\int \frac{\sqrt{x}}{1+\sqrt{x}}$ (let $x = t^2$)

(4) $\int t \tan t^2 \sec t^2\, dt$ (let $t^2 = u$)

(5) $\int t \sec^2(t^2)\, dt$ (let $t^2 = u$)

(6) $\int \frac{1-\sqrt{3x+2}}{1+\sqrt{3x+2}}\, dx$ (let $3x+2 = t^2$)

(7) $\int \frac{dx}{\sqrt{x} - \sqrt[3]{x}}$ (let $\sqrt{x} = t$)

(8) $\int \sqrt{e^x - 1}\, dx$ (let $e^x - 1 = t^2$)

(9) $\int (2x+1)\sqrt{x-5}\, dx$ (let $x-5 = t^2$)

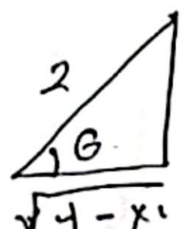
(10) $\int \frac{x^2}{(x-1)^4}\, dx$ (let $(x-1) = t$)

* Trigonometric substitutions

In the case where the integrand contains expressions $\sqrt{a^2-x^2}$, $\sqrt{a^2+x^2}$ & $\sqrt{x^2-a^2}$, $a>0$ ~~we use~~ Trigonometric substitutions are used as followed

ex.	sub.	comments
$\sqrt{a^2-x^2}$	$x = a\sin u$	$\sin^2 u + \cos^2 u = 1$
$\sqrt{a^2+x^2}$	$x = a\tan u$	$1+\tan^2 u = \sec^2 u$
$\sqrt{x^2-a^2}$	$x = a\sec u$	$\sec^2 u - 1 = \tan^2 u$

Ex. Evalute $\int \frac{x^2\,dx}{\sqrt{4-x^2}}$



مثلث التعويضية

Sol.: let $x = 2\sin\theta \rightarrow$

$dx = 2\cos\theta\, d\theta$

$$\Rightarrow \int \frac{x^2}{\sqrt{4-x^2}}\,dx = \int \frac{4\sin^2\theta \cdot \cancel{2\cos\theta}\, d\theta}{\cancel{2\cos\theta}} = 4\int \sin^2\theta\, d\theta$$

$$= 2\int (1-\cos 2\theta)\, d\theta = 2\theta - \sin 2\theta + C$$

من مثلث التعويض

$$= 2\theta - 2\sin\theta\cos\theta + C$$

$$= 2\sin^{-1}\frac{x}{2} - x\cdot\frac{\sqrt{4-x^2}}{2} + C$$

ملاحظة

$\sin 2\theta = 2\sin\theta\cos\theta$

Problems

Evaluate the following integrals

① $\displaystyle\int \frac{\sqrt{9-x^2}}{x}\,dx$ $\boxed{\text{let } x = 3\sin u}$

② $\displaystyle\int \frac{x^2\,dx}{\sqrt{2x-x^2}}$

نعمل إكمال مربع أولاً ،

$2x - x^2 = 1 - (1-x)^2$

$\boxed{\text{let } 1-x = \sin u}$

③ $\displaystyle\int \frac{dx}{x^2\sqrt{4+x^2}}$ $\longrightarrow$ (let $x = 2\tan u$)

④ $\displaystyle\int \frac{x^2\,dx}{\sqrt{x^2-4}}$ (let $x = 2\sec u$)

Integration by Parts : التكامل بالتجزئ

The Rules of integration by Part is

$$\boxed{\int u\,dv = uv - \int v\,du} \quad (*)$$

(i.e) int. by part transfer the integral into simpler integral can be calculated.

Ex.: $\int \frac{x}{\sqrt{1+x}}\,dx$

$\because \int \frac{x}{\sqrt{1+x}}\,dx = \int x(1+x)^{-1/2}\,dx$

let $u = x$ $\qquad dv = (1+x)^{-1/2}\,dx$

$\Rightarrow du = dx$ $\qquad v = 2(1+x)^{1/2}$

$\therefore \int \frac{x}{\sqrt{1+x}}\,dx = 2x(1+x)^{1/2} - 2\int (1+x)^{1/2}\,dx$

$= 2x(1+x)^{1/2} - 2/3\,(1+x)^{3/2} + C$

Ex: $\int x \tan^{-1}x\,dx$

$u = \tan^{-1}x$ $\qquad dv = x\,dx$

$du = \cancel{\frac{1}{\sqrt{\;}}}\frac{1}{1+x^2}\,dx$ $\qquad v = \frac{1}{2}x^2$

$\Rightarrow \int x \tan^{-1}x\,dx = \frac{1}{2}x^2\tan^{-1}x - \frac{1}{2}\int \frac{x^2}{x^2+1}\,dx$

$= \frac{1}{2}x^2\tan^{-1}x - \frac{1}{2}x + \frac{1}{2}\tan^{-1}x + C$ #

Problems

Evaluate the following integrals:-

(1) $\int x e^{-x} \, dx$ ② $\int x^2 \sin x \, dx$

③ $\int \sin^{-1} x \, dx$ ④ $\int \tan^{-1} x \, dx$

⑤ $\int \ln x \, dx$ ⑥ $\int x \sqrt{1+x} \, dx$

⑦ $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} \, dx$ ⑧ $\int x \sin^{-1} x \, dx$

⑨ $\int e^{x} \sin 4x \, dx$ ⑩ $\int \frac{\ln x}{x^3} \, dx$